UniGuide Uganda

AI-Powered University and Course Recommendation System for Students in Uganda

Project Proposal Defence Report

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1. Introduction and problem statement

Thousands of Ugandan students complete the Uganda Advanced Certificate of Education (UACE) each year with limited access to accurate, up-to-date information about university programmes, entry requirements, cut-off points, fees and career prospects. This information is scattered across university websites, changes every academic year, and is hardest to reach for students outside Kampala. As a result, students frequently apply for courses they do not qualify for, or miss courses that match their strengths. UniGuide Uganda is a web-based tool that converts a student's academic results and preferences into a shortlist of programmes they are eligible for, with the reasons for each recommendation.

2. Novelty

What is new. There is no open, reusable dataset of Uganda university admission information. We build one by collecting and verifying official documents, then modelling it in a pipeline that can be re-run each year:

• **Uganda-specific, source-verified data.** The dataset is assembled from the National Council for Higher Education (NCHE) accredited-programme register, official university cut-off-point and fee PDFs, and university programme catalogues. Every programme row is traceable to an official source.

• **Explainable by design.** Recommendations are produced by a rule-based eligibility engine (hard checks plus weighted scoring) augmented by a transparent machine-learning layer. Unlike black-box tools, every output can be justified to a student or counsellor.

• **Local admission constructs.** The system models UACE subject combinations, government versus private cut-off schemes, NCHE minimum entry regulations, and international qualifications handled the NCHE-compliant way.

• **A moving-target data problem.** Cut-off points and fees change every academic year and are published in PDFs. We automate extraction, cleaning and validation instead of hard-coding values that would expire.

Why it matters. A wrong course choice can cost a student years of study and significant fees. A verified, explainable and annually refreshable tool directly reduces that risk, and gives upcountry students the same information quality as those in Kampala.

3. Dataset

3.1 Source

The data comes from three public, official families of documents: (1) the NCHE accredited-programme register and accredited-institutions list; (2) official university cut-off-point and fee publications, including the PUJAB government cut-off schedules; and (3) university programme catalogue pages (Busitema, Gulu, Kyambogo, MUST, Nkumba, Uganda Christian University, Uganda Martyrs University).

3.2 Contents

The dataset records, for each programme: the programme name, the awarding institution, the qualification level, the institution type (public/private), the NCHE tier discipline, accreditation and review dates, and the district. For the curated subset it also records admission details: essential and relevant subjects, minimum grades, principal passes, government and private cut-off points, tuition fees, the academic year and an official source URL.

3.3 How it was collected

Official documents were downloaded from public sources and processed by an automated pipeline. Tabular documents (CSV/Excel) are read directly; PDFs are parsed with table and text extraction to recover cut-off points and fee figures. Because the source data is messy and repeats across files, the pipeline cleans, de-duplicates, validates and merges the records, and produces a single training dataset plus a quality report. The source files are never modified; all outputs are written to a timestamped processed folder and mirrored to `data/cleaned/` for easy access.

4. Features

The classification model uses four inputs. Text features are converted with TF-IDF word n-grams (1-2) over the combined programme and institution text; the categorical inputs are one-hot encoded.

Feature	Type	Meaning
programme	Text	Programme/course name - the main signal of the field of study
institution	Categorical	Awarding university or college
qualification_level	Categorical	Certificate, Diploma, Bachelor, Masters, etc.
institution_type	Categorical	Public or private awarding body
district (context)	Categorical	Location, used by the application for regional filtering

5. Number of records

Dataset	Records
Model training dataset	3,622 records, 12 classes
NCHE accredited programmes (clean)	3,622 (from 3,753 raw)
NCHE accredited institutions (clean)	153 (from 178 raw)
University catalogue entries (clean)	376
Curated seed programmes	75 across 11 universities
Cut-off rows extracted from PDFs	1,361

Career paths	21
Fee rows extracted (incomplete - OCR)	3

6. Target variable

The target is **discipline** - the NCHE Tier Discipline of a programme. This is a supervised multi-class classification problem with 12 classes. The table below shows the distribution of records per class in the training dataset.

Discipline	Records	Discipline	Records
Business	827	Engineering	192
Social and Behavioral Sciences	582	Life Sciences	188
Education	514	Unclassified	103
Physical Sciences and Mathematics	504	Law	43
Medicine and Health Sciences	388	Architecture	17
Arts and Humanities	262	STEM	2

Important. This model predicts a programme's field, not whether a student will be admitted. The dataset contains no student outcome labels, so no admission-success model is claimed.

7. Testing and evaluation

7.1 Data split

The training data is split in a stratified 60/20/20 ratio into 2,172 training, 725 validation and 725 test records. Stratification preserves the class balance of the 12 disciplines across all three sets.

7.2 Models

Three models are trained on identical features and compared: Logistic Regression, Decision Tree, and Random Forest. The model with the best validation macro-F1 is selected and then re-scored once on the held-out test set, which is never used for model selection.

7.3 Metrics

Performance is judged with accuracy, macro-averaged precision, recall and F1, and a confusion matrix. Macro-F1 is the headline metric because the classes are imbalanced, so a model that ignores small classes would still score well on accuracy alone.

7.4 Results

Model	Accuracy	Macro-F1
Logistic Regression (selected)	0.873	0.709

The content-based recommender (TF-IDF cosine similarity over programme and institution text) is evaluated with Precision@5, Recall@5 and NDCG@5 using career-to-programme links as the relevance signal. These values (Precision@5 = 0.1143, Recall@5 = 0.0562, NDCG@5 = 0.1029) are a weak, self-referential proxy, not held-out ground truth, and are reported with that caveat.

Reproducibility: all splits use a fixed random seed (42), so results are deterministic and can be re-run with `python -m ml.train`.

8. Procedures performed

- **Inventory.** All 81 source files were catalogued and classified (read-only) to decide their role.
- **Extraction.** PDFs were parsed for cut-off-point tables and fee text; CSV/Excel were read directly.
- **Cleaning.** Columns were standardised; text, money, cut-off-pair and duration fields were parsed; invalid records were quarantined (not deleted).
- **De-duplication.** Exact and entity-level duplicates were removed with a full audit trail; genuine conflicts were retained and flagged.
- **Merging and validation.** Datasets were joined with blow-up detection; quality checks and a readiness score were produced.
- **Publishing.** Outputs were written to a timestamped run folder and mirrored to `data/cleaned/`.
- **Modelling.** Features were built, three models compared, and the best saved with a model card.
- **Testing.** An automated test suite covers the pipeline, the models and the admin interface.

9. Limitations

- No admission-outcome labels exist, so only programme classification and similarity are modelled.
- Three scanned, image-only PDFs could not be parsed (no OCR engine available) and are flagged for review.
- Fee extraction is incomplete because some source PDFs contain corrupted text; low-yield documents are flagged rather than emitting unreliable values.
- Some classes are very small (e.g. STEM with 2 records, Architecture with 17), which limits their per-class performance.

10. Conclusion

UniGuide Uganda proposes a verified, explainable and annually refreshable recommendation system for Uganda university admissions. This report documents the dataset, its features, its size, the prediction target, and a rigorous testing plan whose measured results are transparently reported.